e61 fiscal model — local research prototype

Run from `projections`:

```
& 'C:\Program Files\R\R-4.5.1\bin\Rscript.exe' --vanilla 'e61 fiscal model\run_app.R'
```

Open http://127.0.0.1:7461. The app binds only to this computer. Set `E61_FISCAL_PORT` to choose another port. Required packages: shiny, bslib, ggplot2. The engine itself uses base R only. No downloads occur when the app runs.

Scope

This is a new, independent scenario engine, not a fitted aggregate regression or an overlay of existing projection outputs. It implements all four requested controls: labour productivity, NGDP growth, government service prices and terms of trade. It also offers two illustrative benefit/service conventions, a simple tax-indexation sensitivity, pension coverage attrition and an interest-rate setting.

It demonstrates the modelling architecture. It is **not yet a validated realistic alternative to PBO/IGR**, a complete current-policy simulation, or an official e61 forecast. Programme eligibility, historical validation and an updated official data vintage are still needed before public release.

Starting data and calibration

`scripts/prepare_inputs.R` copies four existing project inputs into `data/`, with a source-path and checksum manifest. Parent models and files are not changed. The snapshot was prepared on 25 October 2026; this is not the release date of the underlying sources.

- `official_pbo_nfo_wide.csv`: existing consolidated **2025–26** National Fiscal Outlook extract, with forecasts through FY2029–30. This intentionally preserves the existing comparison anchor; it is not an updated 2026–27 PBO extract.
- `population_projection_age_shares.csv`: existing annual population and age-group projection to 2066. Broad bands and interpolation are inherited from the parent preparation workflow; no new demographic forecast is estimated here.
- `historical_gfs_expenses_by_purpose.csv` and `historical_gfs_operating_statement.csv`: existing ABS GFS extracts, 2024–25 composition and conventional interest, with source amounts converted from AUD millions to billions.
- `external_anchors.csv`: manually recorded Age Pension outlay of $62.2bn and 2.67m June recipients from the ANAO audit. Cash outlays are used as an approximate allocation within GFS Social Protection, not added on top of it. The recipient stock is documented but the model projects exposure indices, not recipient counts or means-test eligibility.

2025 primary GFS spending is split into Health, Education, Age Pension, Other Social Protection, Other Public Services and Defence. Conventional interest is removed exactly once. Programme exposure and cost paths carry those allocations to 2030 under the default assumptions; one common factor then reconciles their sum to the PBO's 2030 primary-expense total. These are **e61 allocation estimates**, not official PBO programme forecasts. This calibration is frozen before scenarios are evaluated. Scenario changes start in FY2030–31.

The first version retains aged care and NDIS inside residual Social Protection. They require separate programme data and eligibility/usage mechanisms in the next development stage. Pension means testing and superannuation maturation are represented only by an explicit, adjustable coverage-decline proxy. The source vintages and broad programme mapping must be reconciled before publication.

Economic engine

Let t be years since 2030. Labour volume is population times age shares weighted by fixed propensities: ages 15–34: .65; 35–54: .80; 55–64: .60; 65–74: .20; 75+: .03; children: zero. These are **illustrative employment/hour weights**, not estimated participation projections. Real output is normalised to 2030:

```
productivity index A[t] = (1 + productivity growth)^t
real output Q[t] = (labour volume[t] / labour volume[2030]) * A[t]
terms of trade T[t] = (1 + chosen level change)^min(t/10, 1)
```

With linked NGDP, domestic prices P grow by 2.5% per year:

```
NGDP[t] = NGDP[2030] * Q[t] * P[t] * T[t]^0.20
```

The 0.20 terms-of-trade income elasticity is an exposed judgement. This is a reduced-form nominal income effect, not a full trade model. A positive terms-of- trade shock increases export prices/income without increasing real production.

With the NGDP override, NGDP instead grows at the chosen rate and domestic prices are the residual `P = NGDP / (NGDP[2030] * Q * T^0.20)`. Thus productivity and NGDP are jointly consistent. Higher productivity at fixed NGDP reduces implied inflation; it is not the same experiment as higher productivity at fixed inflation. The implied domestic inflation is reported. CPI, the wage deflator and domestic prices share this price proxy initially. The GDP deflator includes the ToT income effect and is distinct from that domestic proxy. Extreme joint settings can imply deflation: it is shown rather than silently clipped. Productivity can be negative.

Spending

```
spending = calibrated base * demographic exposure * unit cost * coverage * intensity
service unit cost = P * A^wage_pass_through * (1 + extra service price growth)^t
```

The wage pass-through parameter represents net exposure to economy-wide wage growth after assumed service-delivery productivity; it is not separately estimated. Different sector wage/pass-through and delivery-productivity channels should be separated in the next version. The extra service price slider is an additional premium above this baseline, not total relative-price growth. `data/programme_parameters.csv` exposes each age profile, pass-through and excess intensity assumption. Health has 0.5% annual excess intensity growth; Other Social Protection 0.3%; the rest zero. These are illustrative assumptions, not estimates.

- Indexed rules: Age Pension payments follow wages; demographic coverage falls 0.5% annually by default, as a proxy for retirement-income maturation. Other social protection uses a .65 wage pass-through. This is not a full simulation of legislated indexation, eligibility or means tests.
- Constant relative standards: pension coverage stays fixed, and Other Social Protection follows real wages fully. Service quantity/intensity assumptions otherwise match the indexed case; non-zero excess intensity still applies.

- Defence converges from its calibrated share to 2.5% GDP by 2035.
- Net capital investment grows with population, domestic prices and half of productivity growth. It is separate from expenses; depreciation already within GFS expenses is not added again.

Revenue and balance sheet

2030 aggregate revenue exactly matches PBO. Interest receipts are allocated using their share of 2025 GFS revenue. Remaining revenue is split into **illustrative** labour 42%, company 20%, consumption 25%, other 10%, and resource-linked 3% blocks. These weights are not claimed to be observed tax-head shares.

Labour revenue follows labour volume * domestic prices * productivity. Partial bracket creep raises its productivity exponent from 1 to 1.1, with no unbounded nominal-threshold compounding. Company and other revenue follow NGDP; consumption follows domestic nominal output; resource revenue also responds to ToT with an illustrative elasticity of 2. There is no tax microsimulation or revenue cap.

Interest uses opening **gross** debt. The effective rate starts at the rate implied by the 2030 PBO interest bill and 2029 gross debt, converging by 2040 to 4.4% by default. Interest receipts use opening financial assets and an anchor- calibrated yield. Asset purchases maintain the starting financial asset stock per person in domestic purchasing power. Their financing is explicit:

```
fiscal balance = revenue - expenses - net capital investment
gross debt[t] = gross debt[t-1] - fiscal balance[t] + financial asset acquisition[t]
net debt = gross debt - financial assets
```

After gross debt is repaid, further surpluses accumulate in interest-bearing assets. There is no hidden debt floor applied to net debt and no automatic fiscal repair. No maturity buckets, asset valuation changes, inflation-indexed debt or cyclical stabilisation are modelled. The common 2030 official anchor may contain stock-flow adjustments not reconstructible from the rounded published ratios; the new recursive stock-flow identities apply from 2031.

Comparisons and outputs

The reference uses productivity 1.2%, linked NGDP, 0.3% extra service prices and unchanged ToT. It shares the user's selected policy/financing assumptions, so the economic-control comparison is conditional on those settings. Reset returns all settings to defaults. An exact four-factor Shapley calculation allocates economic control interactions in 2066 net debt; it is accounting attribution, not causality.

Download annual reference/scenario results and the assumptions/calibration from the Method & Data tab. `outputs/` holds the default projection, programme calibration, validation results and R session information. No probability bands are shown: structural and policy uncertainty have not yet been quantified.

Run validation from `projections`:

```
& 'C:\Program Files\R\R-4.5.1\bin\Rscript.exe' --vanilla 'e61 fiscal model\tests\validate_model.R'
```

Checks cover aggregate/anchor reconciliation, no duplicate pension spending, stock-flow identities, interest bases, surplus asset accumulation, controlled productivity/NGDP/price/ToT changes and attribution interactions. They establish implementation consistency, not predictive accuracy. Historical backtesting and programme validation remain necessary.

On 25 October 2026 all 33 engine checks passed. An optional real-Chrome smoke test (`tests/browser_smoke.R`, requires chromote) also verified each of the four controls independently, all tabs, reset, the live Shiny connection and both downloads. Browser screenshots are saved as `outputs/app_reference.png` and `outputs/app_scenario.png`; detailed results are in `outputs/browser_smoke_results.txt`. The running process ID is recorded in `outputs/shiny_pid.txt`, with startup messages in `outputs/shiny_stderr.log`.

Next development priorities

1. Update and reconcile PBO, ABS, population and 2026 IGR vintages; distinguish Commonwealth/state/local coverage and eliminate internal transfers.
2. Replace proxy age exposures with participation/hours projections and sourced service-use and age-cost profiles.
3. Separate NDIS and aged care; improve pension eligibility, indexation, means testing and retirement-income maturation rather than fitting coverage attrition.
4. Estimate tax-head shares and elasticities; add explicit threshold rules.
5. Backcast components and balances and compare historical forecast vintages; evaluate structural uncertainty against the existing top-down challenger.
6. Review assumptions and sensitivity ranges before any public deployment.